Dr Yidan Xue

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Education

DPhil Engineering Science, *University of Oxford* Jan 2023
Thesis: Modelling oxygen transport and tissue damage in the human brain.
Supervisor: Professor Stephen Payne. Submission: Aug 2022/Viva: Oct 2022.

BEng (Hons) Mechanical Engineering, *The University of Edinburgh* Jul 2019
Dissertation: Computational simulation and validation of flows in branching blood vessels.
Supervisors: Drs Dong-hyuk Shin and Rudolf Hellmuth. First Class Honours.
I completed the first two years of my undergraduate study (2015–2017) at Xiamen University, China.

Employment

BHF/UK CEiRSI Transition Fellow, *Department of Computer Science, The University of Manchester* Apr 2026 – Present
Research Associate, *School of Health Sciences, The University of Manchester* Aug 2024 – Mar 2026
Research Associate, *School of Mathematics, Cardiff University* Jan 2024 – Jul 2024
EPSRC Postdoctoral Research Associate, *Mathematical Institute, University of Oxford* Oct 2022 – Dec 2023
Retained Lecturer in Mathematics, *Jesus College, University of Oxford* Jan 2023 – Sep 2023
Research Intern, *Institute of Mechanics, Chinese Academy of Sciences* Jul 2018 – Sep 2018
Research Intern, *School of Engineering, The University of Edinburgh* May 2018 – Jul 2018

Publications

Journal Articles:

1. Lin F, Zakeri A, **Xue Y**, MacRaild M, Dou H, Zhou Z, Zou Z, Sarrami-Foroushani A, Duan J, Frangi AF. From Pixels to Polygons: A Survey of Deep Learning Approaches for Medical Image-to-Mesh Reconstruction. *Medical Image Analysis*. 2026.
2. Jabi W, **Xue Y**, Woolley TE, Kaouri K. 3D Topological Modeling and Multi-Agent Movement Simulation for Viral Infection Risk Analysis. *Architectural Science Review*. 2025.
3. Mao Y, Liu Y, Zhai M, Jin P, Li W, Dong X, Chen F, Wang X, Wang Y, Zhang G, Li H, Yang Y, Zhang H, Liu J, Guo Y, Wu Y, **Xue Y**, Zhang J‡, Frangi AF‡, Yang J‡. Precision TAVR Quantification—AI-accelerated TAVR Planning Reduces Assessment Time by 80% in Bicuspid Aortic Stenosis. *European Heart Journal – Imaging Methods and Practice*. 2025;3(4):qyaf153. ‡: corresponding authors.
4. Payne SJ, **Xue Y**, Kuo J-F, El-Bouri WK. Transit time mean and variance are markers of vascular network structure, wall shear stress distribution and oxygen extraction fraction. *Biomechanics and Modeling in Mechanobiology*. 2025;24:1155–1167.
5. **Xue Y**. Computing Stokes flows in periodic channels via rational approximation. *Proceedings of the Royal Society A*. 2025;481:20240676.
6. **Xue Y**, Payne SJ, Waters SL. Stokes flows in a two-dimensional bifurcation. *Royal Society Open Science*. 2025;12:241392.

7. **Xue Y**, Jabi W, Woolley TE, Kaouri K. Modelling indoor airborne transmission combining architectural design and people movement using the VIRIS simulator and web app. *Scientific Reports*. 2024;14:28220.
8. **Xue Y**, Waters SL, Trefethen LN. Computation of two-dimensional Stokes flows via lightning and AAA rational approximation. *SIAM Journal on Scientific Computing*. 2024;46(2):A1214–A1234. **[ESI Highly Cited Paper, SIAM Reproducibility Badge]**.
9. **Xue Y**[‡], Georgakopoulou T[‡], van der Wijk A-E, Józsa TI, van Bavel E[‡], Payne SJ[‡]. Quantification of hypoxic regions distant from occlusions in cerebral penetrating arteriole trees. *PLOS Computational Biology*. 2022;18(8):e1010166. ‡: co-first/co-senior authors.
10. Miller C, Padmos RM, van der Kolk M, Józsa TI, Samuels N, **Xue Y**, Payne SJ, Hoekstra AG. In silico trials for treatment of acute ischemic stroke: design and implementation. *Computers in Biology and Medicine*. 2021;137:104802.
11. **Xue Y**, El-Bouri WK, Józsa TI, Payne SJ. Modelling the effects of cerebral microthrombi on tissue oxygenation and cell death. *Journal of Biomechanics*. 2021;127:110705. **[Special Issue on Thrombus Mechanics]**.
12. **Xue Y**, Hellmuth R, Shin D-H. Formation of vortices in idealised branching vessels: a CFD benchmark study. *Cardiovascular Engineering and Technology*. 2020;11(5):544–559. **[Cover Image]**.

Submitted Articles/Preprints:

13. Padmos RM, Józsa TI, **Xue Y**, Payne SJ, Hoekstra AG. A Multi-Scale Model for Perfusion-Based Infarct Estimation in Acute Ischaemic Stroke Patients. 2023. submitted.
14. Miller C, Padmos R, Konduri P, Józsa TI, **Xue Y**, Arrarte Terreros N, van der Kolk M, Payne SJ, Marquering H, Majoie C, Hoekstra A. In silico trials of acute ischemic stroke: predicting the total potential for improvement to patient functional outcomes. 2025. submitted. Also available on arXiv: `https://doi.org/10.48550/arXiv.2511.02088`.
15. Zhou Z, Zakeri A, Dou H, **Xue Y**, MacRaild M, Huang J, Lin F, Sarrami-Foroushani A, Duan J, Frangi AF. Synthetic Anatomy: Deep Learning Models for Virtual Population Generation—A Review. 2025. submitted. Also available on medRxiv: `https://doi.org/10.1101/2025.10.28.25338782`.
16. **Xue Y**, Sarrami-Foroushani A, Kreuzer S, McVeigh C, Grumbridge M, Bombien R, McLaren A, Pandey PK, Ademiloye AS, Tyler P, MacRaild M, Zakeri A, Revell A, Banda G, Hill D, Nithiarasu P, Manfrin A, De Cunha Maluf-Burgman M[‡], Frangi AF[‡]. A Consensus Framework for In-Silico Regulatory Evaluation of High-Risk Medical Device Design: A Transcatheter Aortic Valve Implantation Exemplar. 2026. submitted. ‡: co-senior authors.
17. Mao Y[‡], Wei H[‡], Liu Y[‡], Lu F, Wu Y, **Xue Y**, Zakeri A, Sarrami-Foroushani A, Li Y, Song L[‡], Qiao Y[‡], Frangi AF[‡], Yang J[‡]. Automatic Quantification of Aortic Valve Fibrocalcification for Clinical Phenotyping and Prognostic Stratification. 2026. submitted. ‡: co-first/corresponding authors.

DPhil Thesis:

18. **Xue Y**. Modelling oxygen transport and tissue damage in the human brain. 2022. DPhil thesis.

Invited Talks

1. *Towards StressMAP: A TAVI deployment simulation workflow for stress-based surrogate modelling of pacemaker dependency*, The 10th Biennial Heart Valve Biology & Tissue Engineering Meeting, The Royal Society, London, UK, Sep 2025
2. *Mechanistic simulations in real-world systems for medical device innovation*, Simulation Workshop

(5-min lightning talk), AI for Research: How Can AI Disrupt the Research Process, The University of Manchester, UK, Jun 2025

3. *Computation of 2D Stokes flows via lightning and AAA rational approximation*, Physical and Applied Mathematics Seminar, The University of Manchester, UK, Nov 2024
4. *Modelling physiological flows and transport at low Reynolds numbers*, CIMIM Seminar (inaugural talk), The University of Manchester, UK, Oct 2024
5. *A state-of-the-art epidemic simulator and web app for viral transmission in indoor spaces*, SIAM Conference on the Life Sciences, Portland, Oregon, US, Jun 2024
6. *Computation of two-dimensional Stokes flows via lightning and AAA rational approximation*, Computational and Applied Math Seminar, Peking University, China, May 2024
7. *Computation of physiological flows and transport at low Reynolds numbers*, Applied and Computational Mathematics Seminar, Cardiff University, UK, Feb 2024
8. *Computation of 2D Stokes flows via lightning and AAA rational approximation*, Numerical Analysis Group Internal Seminar, University of Oxford, UK, May 2023
9. *Modelling oxygen transport in the human cerebral microvasculature*, British Applied Mathematics Colloquium, Bristol, UK, Apr 2023

Selected Contributed Talks

10. *An automatic workflow for in-silico clinical trials of TAVI devices: From images to deployment and stress analysis*, 2025 MDIC Symposium on Computational Modeling & Simulation, College Park, Maryland, US, Nov 2025
11. *Computation of 2D Stokes flows via lightning and AAA rational approximation*, Numerical Analysis in the 21st Century in honour of Nick Trefethen's retirement from Oxford, Oxford, UK, Aug 2023
12. *Modelling human cerebral tissue damage caused by acute ischaemic stroke*, 9th World Congress of Biomechanics (WCB), Taipei (online), Jul 2022

Teaching

University of Oxford, Mathematical Institute/Department of Engineering Science

One contact hour requires at least two hours of preparation and marking at Oxford.

Tutor, A1 Differential Equations 1, Oriel College, class size: 1–2, contact hours: 8	Fall 2023
Tutor, A7 Numerical Analysis, Jesus College, class size: 1–2, contact hours: 15	Spring 2023
Tutor, C5.6 Applied Complex Variables, MI, class size: 10–12, contact hours: 16	Spring 2023
Lead Tutor, B17 Biomechanics, EngSci, class size: 3–4, contact hours: 13	Spring 2022

Mentoring

The University of Manchester

Siru Zhou, MSc Health Data Science	2026
Peixuan Guo, MSc Health Data Science	2026
Chen Zheng, MSc Health Data Science	2026

University of Oxford

Benjamin Nicholls-Mindlin, MSc Mathematical Modelling and Scientific Computing	2023

Rational Stokes Methods for Tissue Engineering Applications, co-supervised with Professors Sarah Waters and Helen Byrne. The thesis received the second highest distinction.

Funding

(2026) **BHF Manchester Centre of Research Excellence Transition Fellowship**, *The British Heart Foundation (BHF) Manchester Centre of Research Excellence (CRE) & UK CEiRSI*. Digital twins to unravel transcatheter heart valves mechanics in bicuspid patients. Two-year fellowship.
(2024) **UK RS&IN Implementation Phase: Human Health (CERSIs)**, *Innovate UK & MRC*, UK CEiRSI - The UK's Centre of Excellence on In-silico Regulatory Science and Innovation - Pilot Phase, Technical Committee Member & Academic Rapporteur of Pilot 4. PI: Professor Alejandro Frangi.
(2022) **EPSRC Postdoctoral Research Associate**, *University of Oxford MPLS Division & EPSRC*. Modelling microthrombi distribution at vessel bifurcations. One-year postdoctoral position awarded to up to 5 DPhil graduates from MPLS Division.
(2018) **Summer Research Scholarship**, *School of Engineering, The University of Edinburgh*. Flow simulation of fuel injection in internal combustion engines. Two-month research project.

Awards

(2019) **IMechE Best Student Prize**, The University of Edinburgh.
(2018) **3rd Year Class Medal for Mechanical Engineering**, The University of Edinburgh.
(2018) **Edinburgh Award**, The University of Edinburgh.
(2017/2018) **2+2 Student Scholarships**, The University of Edinburgh.
(2016) **1st Prize Scholarship for Academic Excellence**, Xiamen University.

Media

SIAM News (June 13, 2024) Epidemic Simulator and Web App Models Viral Transmission in Indoor Spaces.

Memberships

EMS Topical Activity Group Mathematical Modelling in Life Sciences, *Member*	2026 – Present
ISO TC/150 Implants for surgery, *Committee Member*	2026 – Present
BSI CH/150 Implants for surgery, *Committee Member*	2026 – Present
BSI Young Professionals Network, *Member*	2025 – Present
Society for Industrial and Applied Mathematics (SIAM), *Member*	2024 – Present
VPH Institute, *Member*	2020 – 2021, 2024 – Present
European Society of Biomechanics, *Member*	2023 – 2024

Service and Outreach

Outreach/Public Engagement:

Pint of Science (Manchester), *Organiser (with 4 others)* — 2024 – 2025
'Tech Me Out' events: *'From Virtual Patients to Real Solutions: Medical Innovation on Tap'*, *'Learning to Decommission: Robots in the Nuclear World'* and *'The Enzyme Engineers: Crafting Life's Tiny Machines'*, 12 speakers, more than 160 attendees.

The Welsh Government, *Policy Modelling Group, Member* — 2024 – 2025

Institutional Service:

The University of Manchester:

Christabel Pankhurst Institute, *Fire Marshal* — 2025 – Present
UK CEiRSI, Pilot 4 on TAVI, *Academic Rapporteur* — 2025 – Present

UK CEiRSI, Technical Committee, *Member* 2025 – Present
Recruitment Interview Panel, *Member* 2024

Cardiff University:
Vendor Selection Panel, *Member* 2024

University of Oxford:
Undergraduate Admissions Panel (Maths & Stats at Oriel College), *Member* 2023
Disability Advisory Service, *Non-Medical Support Worker* 2019 – 2020

Conference Organisation and Leadership

WCCM-ECCOMAS 2026, Advanced Physics-based and Data-driven Modeling for Biomedical Digital Twin (minisymposium), *Organiser (with Dongwei Ye and Elena Zappon)*, Munich, Germany, Jul 2026 (incoming)
UK CEiRSI In Silico Regulatory Airlock Panel (Pilot 4), In-Silico Trials of TAVI Devices: Blood Flow Models Predicting Aortic Stresses and Valve Leakage, *Lead (with Martha De Cunha Maluf-Burgman and Ali Sarrami-Foroushani)*, Manchester (online), UK, Nov 2025
The 6th China-UK Forum of Young Scholars in Manchester "AI Applications Across Disciplines", *Invited Guest and Panel Judge*, Manchester (online), UK, Jun 2025
Virtual Imaging Trials in Medicine 2025, Innovative Computational Techniques in Medical Imaging, *Session Chair*, Manchester, UK, Jun 2025
Numerical Analysis in the 21st Century Conference, in honour of Nick Trefethen's retirement from Oxford, Numerical Methods for Differential Equations, *Session Chair*, Oxford, UK, Aug 2023

Journal Article Reviewer

Applied & Computational Mathematics: Computational Mechanics, IMA Journal of Numerical Analysis, International Journal of Computer Mathematics, Journal of Computational Physics, Journal of Fluid Mechanics, PLOS Computational Biology, The ANZIAM Journal

Engineering: Biotechnology and Bioengineering, Building Simulation, Cardiovascular Engineering and Technology

Computer Science & AI: IEEE Transactions on Pattern Analysis and Machine Intelligence, Journal of Open Source Software

Interdisciplinary: Journal of the Royal Society Interface

Referees

Professor Alejandro Frangi, School of Health Sciences/Department of Computer Science, The University of Manchester, a.frangi@manchester.ac.uk
Professor Stephen Payne, School of Engineering Mathematics and Technology, University of Bristol, stephen.payne@bristol.ac.uk
Professor Nick Trefethen, School of Engineering and Applied Sciences, Harvard University, trefethen@seas.harvard.edu
Professor Sarah Waters, Mathematical Institute, University of Oxford, waters@maths.ox.ac.uk